WHITEPAPER

FinOps Response Portal MVP

Contents

Purpose

This MVP defines the smallest useful version of the AI-assisted FinOps response workflow.

The goal is not to build a broad FinOps agent. The goal is to remove the slowest, most manual part of the current process:

- FinOps sends a report item to the owning team.
- The team needs enough context to respond.
- The response must be structured, auditable, and dashboard-ready.
- Reassignment and disputes must be captured without mutating catalogues or AWS resources.

The MVP delivers one controlled response workflow for one report feed, one approved model, one Teams bot conversation interface, and S3-backed state/export.

MVP Outcome

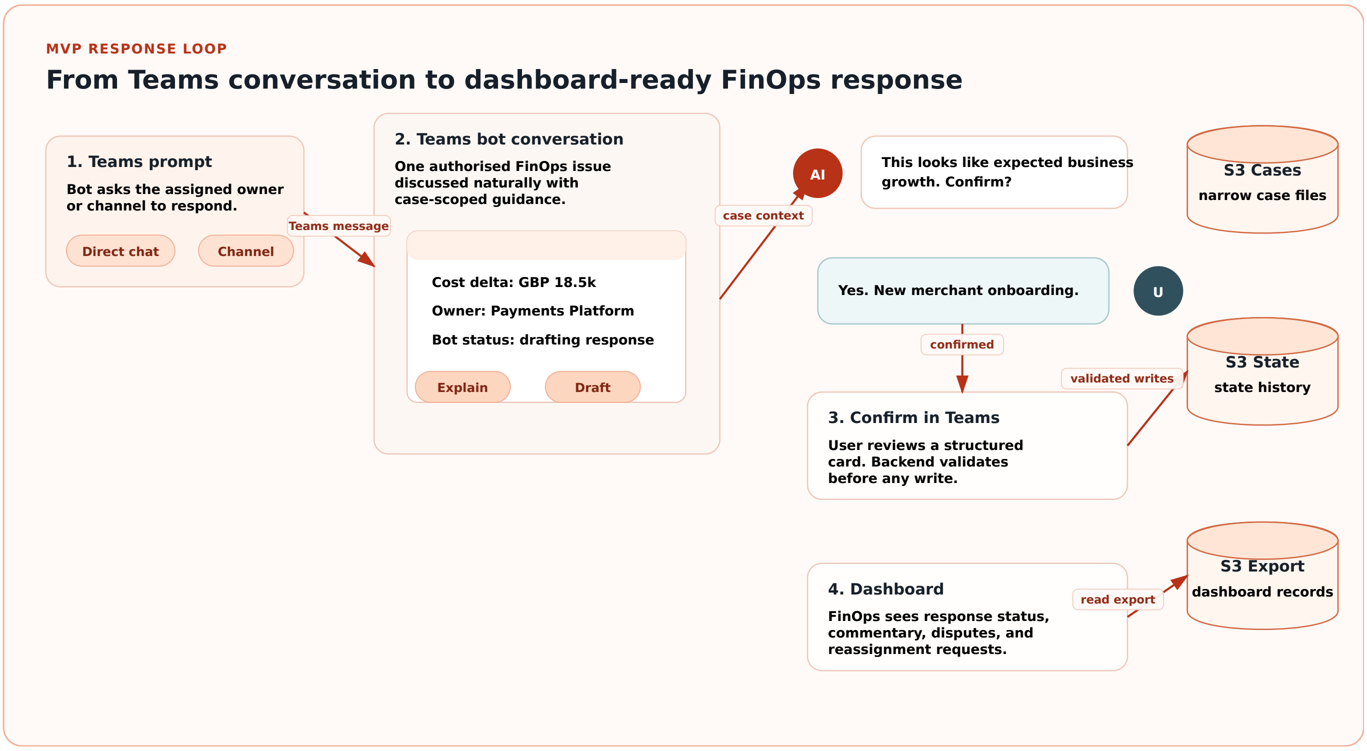
At the end of the MVP, FinOps can ask a user to respond to a specific report item inside Teams. The user discusses the case naturally with a bot, reviews a structured summary inside the conversation, explicitly confirms the response, and the system writes an auditable S3-backed output that dashboards can consume.

The value is:

Value	MVP Mechanism
Faster responses	Case-specific explanation and guided response
Better dashboard commentary	Structured response schema
Lower chasing effort	Status captured centrally
Safer AI adoption	Bedrock only assists; backend owns all writes
Clear audit trail	View, draft, submit, and export events recorded

MVP Visual Overview

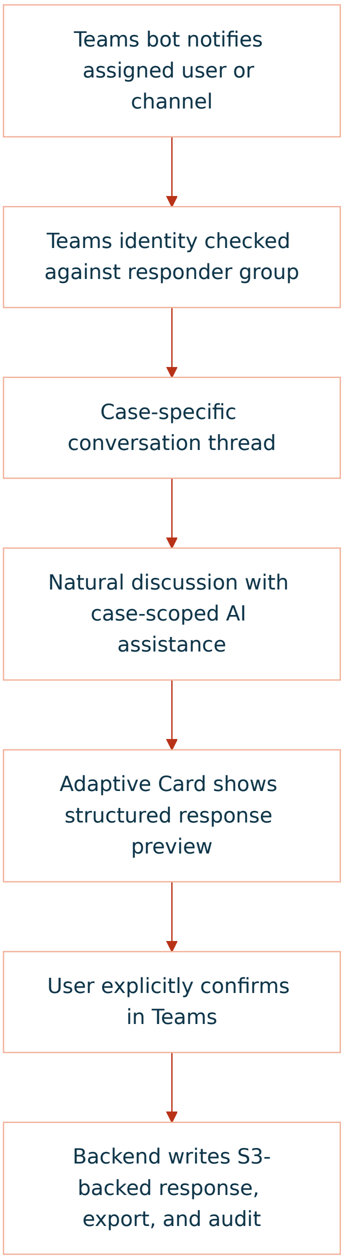
The whole MVP can be understood as a controlled conversational response loop. Users receive a Teams prompt, discuss the assigned FinOps case with a bot, confirm a structured response inside Teams, then the backend writes S3-backed outputs for audit and dashboarding.



Conversation Interface Model

The MVP has one canonical user interface: a Microsoft Teams bot conversation.

The user should be able to complete the MVP without opening a separate web page. The bot can use Teams messages and Adaptive Cards to present the issue, ask clarifying questions, collect the response, and request final confirmation.



Primary Interface: Teams Bot Conversation

The user interacts with one Teams conversation for one report item. That conversation may be a direct chat with the assigned owner or a channel thread where the assigned responder group is authorised to reply.

The bot must make three things obvious immediately:

Question	Bot Answer
What is this issue?	Teams message or Adaptive Card with summary, cost delta, service, period, and current owner
Why did I receive it?	Assigned team/application/product and authorised responder group
What can I do now?	Reply naturally, ask for explanation, or choose one of the controlled response actions

Minimum Teams conversation elements:

Area	Contents
Initial bot message	Issue ID, report period, service, cost delta, status, due date
Assignment context	Product, application, team, responder group
Evidence summary	Source report reference, detected drivers, dashboard reference if present
Natural discussion	User can ask questions and explain the situation in plain language
Bot guidance	Plain-English explanation, follow-up questions, category suggestions
Response capture	Bot maps the conversation to justification, dispute, reassignment, or investigation
Confirmation card	Structured response preview with Confirm, Edit, and Cancel actions
Audit events	View/notify, assist, draft, confirm, submit, and write events

The bot is not a broad FinOps chatbot. Each conversation is scoped to one authorised case unless the user explicitly asks to switch to another assigned case.

User Actions

The user gets four primary actions.

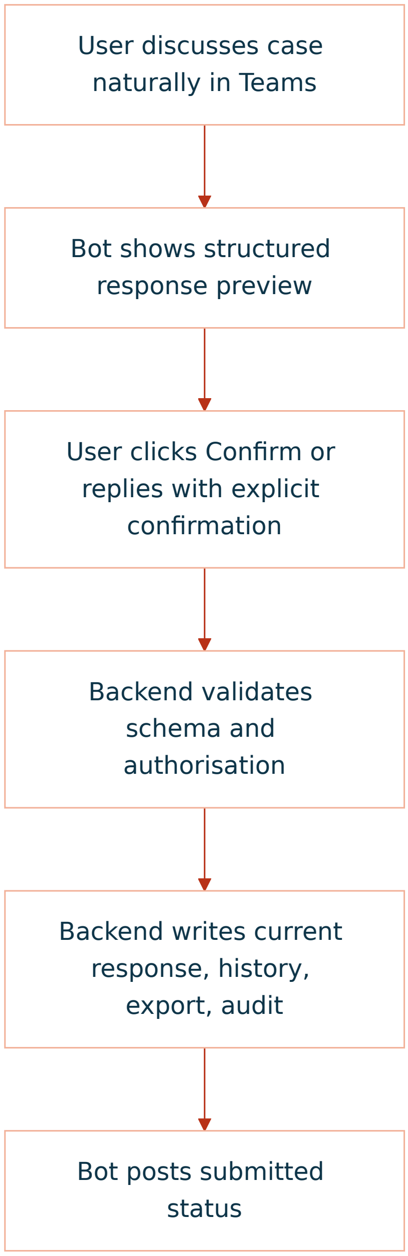
Action	Teams Pattern	Backend Result
Provide justification	User explains naturally; bot drafts structured justification for confirmation	`justified` response written
Dispute issue	User challenges data, amount, service, or allocation; bot captures structured reason	`disputed` response written
Request reassignment	User names proposed owner/application/team; bot captures reassignment request	`reassignment_requested` response written
Mark needs investigation	User says they need time or more evidence; bot captures reason and next step	investigation state written

Secondary actions:

Action	MVP Behaviour
Ask for explanation	AI explains the current case file only
Improve wording	AI drafts clearer finance-friendly wording
Classify response	AI suggests controlled category; user confirms
Attach evidence link	User pastes URL/reference metadata; backend validates shape
Cancel	No response write; audit may record view/session only

Final Submission Flow

Every final response follows the same pattern.



There is no hidden model action after the user confirms. The backend writes only the reviewed structured response.

Email Interface

Email is optional in the MVP and is used for notification or escalation only. It should point the user back to Teams, not to a separate required web UI.

Minimum email content:

Field	Example
Subject	`FinOps response required: Checkout API EC2 increase`
Issue summary	`EC2 spend increased by £18.5k for May 2026`
Assigned owner	`Payments / Checkout API / Payments Platform`
Due date	`2026-06-07`
Action link	`Open Teams conversation`

The email link contains report ID and issue ID only. It does not grant access by itself.

TEXT

```
https://teams.microsoft.com/l/chat/0/0?users=finops-response-bot@company.com&topicName=finops-2026-05-001234
```

Teams Interface

Teams is the MVP interface. The bot should call the same backend API for every read, model assist, draft, validation, confirmation, write, and audit event.

MVP Teams scope:

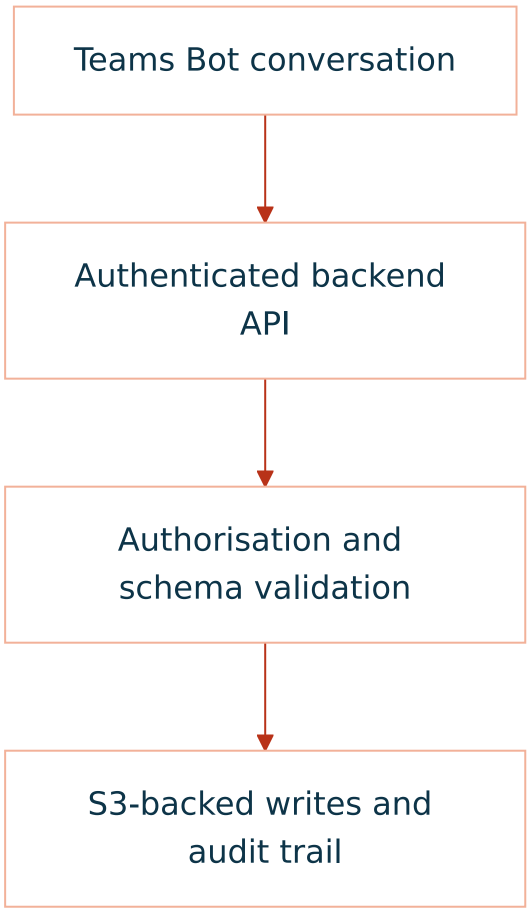
Teams Feature	MVP Use
Bot notification	Tell assigned user or team a response is required
Natural-language chat	Let the user ask questions, explain context, dispute, or request reassignment
Adaptive Card	Show summary, due date, status, suggested category, and confirmation actions
Reminder message	Notify before due date or when status changes
Final confirmation	User confirms the structured response inside Teams

Complex justification, disputes, reassignment, and evidence review should still happen through the bot conversation. If the conversation becomes ambiguous, the bot should ask targeted follow-up questions and require explicit confirmation before submission.

Teams card actions:

Button	Behaviour
Explain	Calls backend; returns case-scoped explanation
Draft response	Converts the conversation into a structured preview
Confirm response	Submits the reviewed structured response
Edit response	Reopens the draft in the Teams conversation
Mark needs investigation	Captures reason and proposed next step
Request reassignment	Captures proposed owner/application/team

The safe pattern is:



Conversation State Model

The bot should expose state clearly in the thread and in any card it posts.

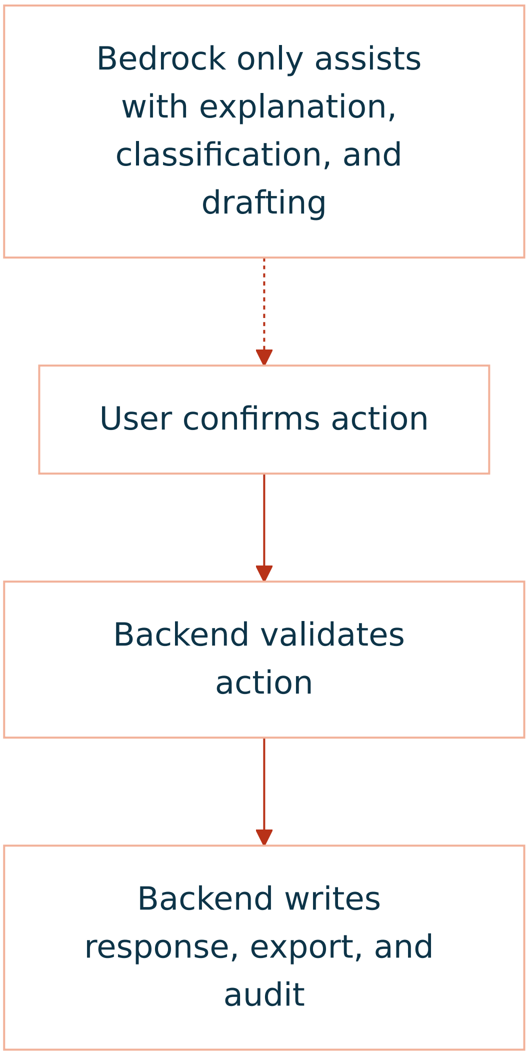
State	Meaning	User Experience
Awaiting response	No final response submitted	Bot has prompted the owner and can answer case questions
In discussion	User has started but not submitted	Bot can continue asking follow-up questions
Awaiting confirmation	Structured draft exists	Bot shows preview and asks for explicit confirmation
Submitted	Final response written	Bot posts submitted summary plus audit reference
Pending FinOps review	Dispute or unclear response needs review	Bot tells the user FinOps review is pending
Reassignment requested	Proposed new owner captured	Bot tells the user reassignment is pending approval
Closed	No further action needed	Bot posts read-only summary if asked

Interface Acceptance Criteria

Requirement	Acceptance Test
User understands why they received the case	Bot message shows current assignment and responder group
User understands the financial issue	Bot message or card shows period, service, cost delta, and drivers
User can act without free-form email	Four controlled response outcomes can be reached from Teams
User can review before submit	Bot shows structured preview before final write
AI assistance is bounded	Bot only references current case context
Teams/email are not security boundaries	Backend validates Teams identity and responder-group membership
Dashboard data is produced	Submit creates dashboard export record
Audit is complete	View, assist, submit, and write events are recorded

Non-Negotiable Constraint

The model is not the system of record and does not write to AWS services.



Smallest Useful Scope

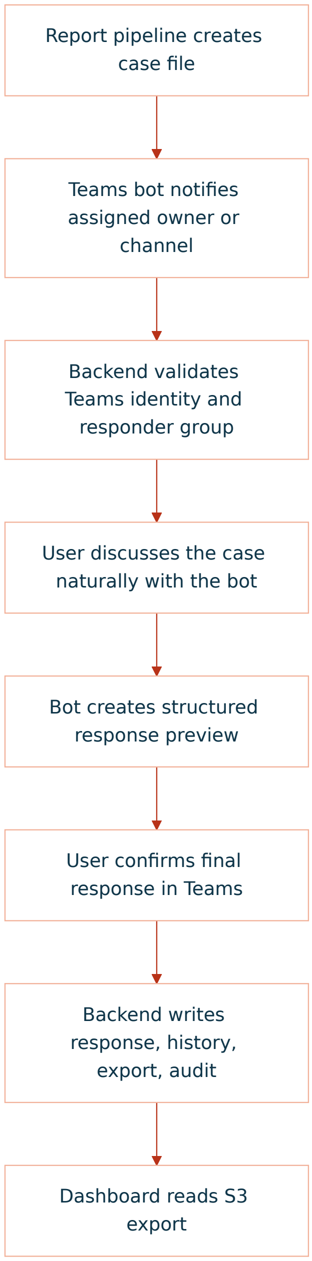
In Scope

Capability	MVP Definition
One report source	A pre-built case file per report item
One primary interface	Teams bot conversation with Adaptive Card checkpoints
Optional email notification	Email can remind or escalate, but Teams remains the response interface
One model	One approved Bedrock text model
Four response types	Justification, dispute, reassignment request, needs investigation
S3-backed storage	Case files, current response, history, export, audit
Dashboard export	One S3 prefix with dashboard-ready JSON or Parquet
Audit events	Views, model assists, submissions, export writes

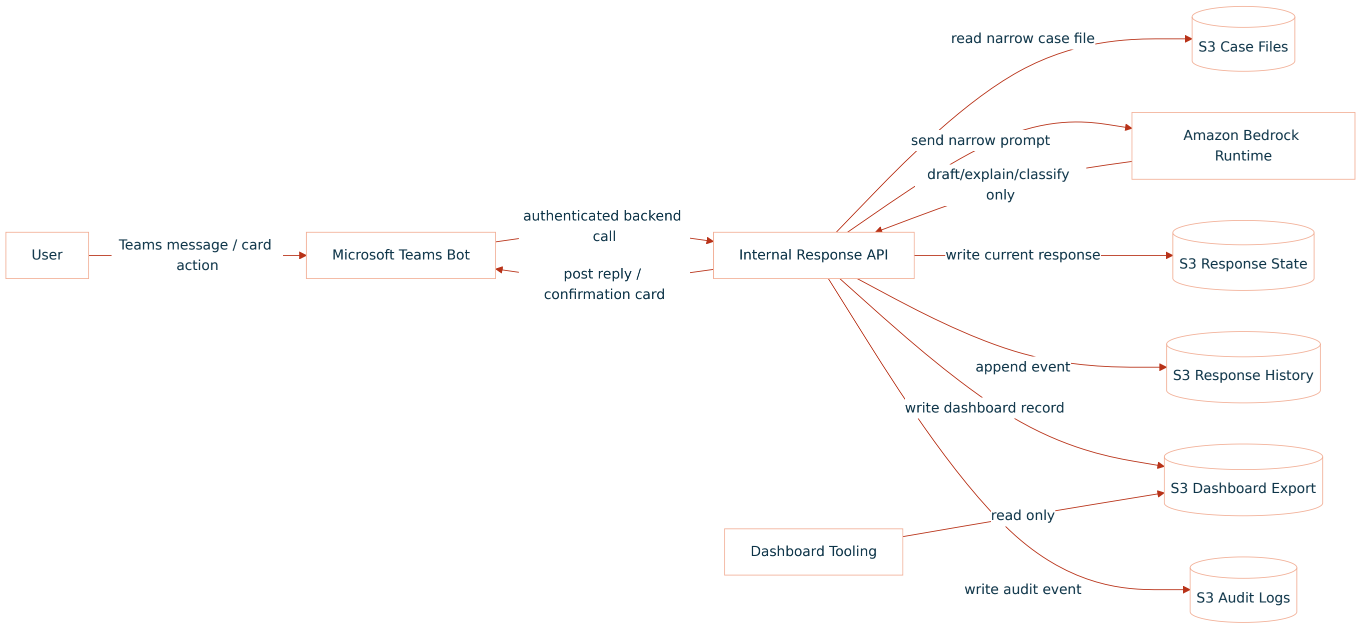
Out of Scope

Excluded	Reason
Autonomous remediation	Not required to capture FinOps responses
Raw CUR access from the app	Case files are generated upstream
Live AWS infrastructure inspection	The app does not diagnose resources directly
IAM, tag, billing, or catalogue mutation	MVP captures requests only
Broad chatbot over all cost data	Interaction is scoped to one case
Dedicated web response UI	Not needed for the MVP; Teams is the user-facing interface
Direct browser-to-S3 writes	Backend must validate and audit all writes
Multiple model providers	One approved model is enough to prove value

MVP User Journey



MVP Architecture



Case File Contract

The upstream report pipeline creates one case file per report item. The MVP app reads this file; it does not query raw CUR directly.

Minimum case file:

JSON

```json
{
  "issue_id": "finops-2026-05-001234",
  "report_id": "monthly-cloud-report-2026-05",
  "report_period": "2026-05",
  "status": "awaiting_response",
  "assigned_team": "Payments Platform",
  "assigned_product": "Payments",
  "assigned_application": "Checkout API",
  "service": "AmazonEC2",
  "cost_delta": 18500.25,
  "currency": "GBP",
  "summary": "EC2 spend increased materially versus prior month.",
  "detected_drivers": [
    "c7g.4xlarge running hours increased",
    "EBS gp3 usage increased",
    "Production autoscaling floor appears higher"
  ],
  "allowed_responder_groups": [
    "payments-platform"
  ],
  "response_due_date": "2026-06-07"
}
```

Response Types

Type	User Intent	Final Status
Justification	Accept the spend and explain why	`justified`
Dispute	Challenge data, amount, service, or allocation	`disputed`
Reassignment request	Suggest a different owner	`reassignment_requested`
Needs investigation	Cannot explain yet	`in_discussion` or `pending_finops_review`

Response Contract

The backend writes one current response and one immutable history event.

```json
{
  "issue_id": "finops-2026-05-001234",
  "response_id": "resp-001",
  "response_type": "justification",
  "status": "justified",
  "submitted_by": "user@company.com",
  "submitted_at": "2026-05-21T14:30:00Z",
  "justification_category": "expected_business_growth",
  "business_reason": "Traffic increased due to new merchant onboarding.",
  "technical_reason": "Additional production capacity was required.",
  "expected_to_continue": true,
  "requires_finops_review": false,
  "ai_summary": "The team accepted the increase as expected business growth."
}
```

Dashboard Export Contract

The dashboard does not need the full transcript. It needs a compact, trusted record.

```json
{
  "report_period": "2026-05",
  "issue_id": "finops-2026-05-001234",
  "product": "Payments",
  "application": "Checkout API",
  "team": "Payments Platform",
  "service": "AmazonEC2",
  "cost_delta": 18500.25,
  "currency": "GBP",
  "status": "justified",
  "response_type": "expected_business_growth",
  "response_summary": "Increase accepted due to merchant onboarding and higher baseline capacity.",
  "submitted_by": "user@company.com",
  "submitted_at": "2026-05-21T14:30:00Z",
  "requires_finops_review": false,
  "requires_catalog_update": false
}
```

S3 Layout

Use one bucket or approved bucket set. Keep prefixes explicit.

```text
s3://<bucket>/case-files/report_period=2026-05/issue_id=finops-2026-05-001234.json
s3://<bucket>/responses-current/report_period=2026-05/issue_id=finops-2026-05-001234.json
s3://<bucket>/responses-history/report_period=2026-05/issue_id=finops-2026-05-001234/event_ts=...json
s3://<bucket>/dashboard-export/report_period=2026-05/team=payments-platform/part-....json
s3://<bucket>/audit/report_period=2026-05/issue_id=finops-2026-05-001234/event_ts=...json
```

Authorisation Rule

```gherkin
User is authenticated
AND user belongs to the assigned responder group OR FinOps admin group
AND issue exists
AND issue is open
AND requested response type is allowed
AND final response passes schema validation
```

Prompt Boundary

The model receives only:

Context	Included
Issue summary	Yes
Current assignment	Yes
Detected drivers	Yes
Allowed response types	Yes
User draft response	Yes
Raw CUR	No
Other teams' issues	No
AWS write credentials	No
Catalogue write access	No

The model may return:

- Plain-English explanation.
- Follow-up question.
- Suggested response category.

- Draft summary.
- Confidence or review flag.

The model may not:

- Submit a response without confirmation.
- Write S3.
- Change assignment directly.
- Claim a root cause without evidence.
- Access unrelated data.

Minimum AWS Ask

Area	MVP Ask
Runtime	One private ECS, Lambda, or approved internal app runtime
Auth	Teams identity mapped to enterprise email and responder-group claims
Bedrock	Invoke one approved model in one approved region
Network	Private route to Bedrock Runtime, S3, KMS, CloudWatch Logs, Secrets Manager if used
S3 read	Case file prefix only
S3 write	Responses current/history, dashboard export, audit prefixes
KMS	Decrypt case files; encrypt/decrypt written objects
Logs	Runtime logs and business audit events

MVP Success Criteria

Criterion	Evidence
User can respond to a case	Submitted response exists in S3 current prefix
Response is structured	JSON schema validation passes
Dashboard can consume output	Export record appears in dashboard prefix
AI is controlled	Bedrock has no direct S3/catalogue/AWS write path
Access is scoped	Unauthorised user cannot view or submit case
Audit is complete	View, model assist, submit, and export events exist
FinOps review is possible	Dispute and reassignment requests are visible

MVP Build Order

Step	Deliverable
1	Case file schema and sample report item
2	Teams bot registration, command surface, and notification/card templates
3	Backend read/authorisation path using Teams identity and responder groups
4	Bedrock explanation and guided conversational drafting
5	Deterministic confirmation and submit endpoint
6	S3 current/history/export/audit writes
7	Dashboard reads export prefix
8	Pilot with one team, one Teams channel, and one report period

Final MVP Statement

Build one Teams bot response workflow for one report feed.

Use Bedrock only to explain, ask, classify, and draft.

Use deterministic backend logic for all validation, authorisation, state changes, S3 writes, and audit.

Publish one dashboard-ready S3 export.

Do not remediate, mutate catalogues, inspect live infrastructure, or expose broad cost data.